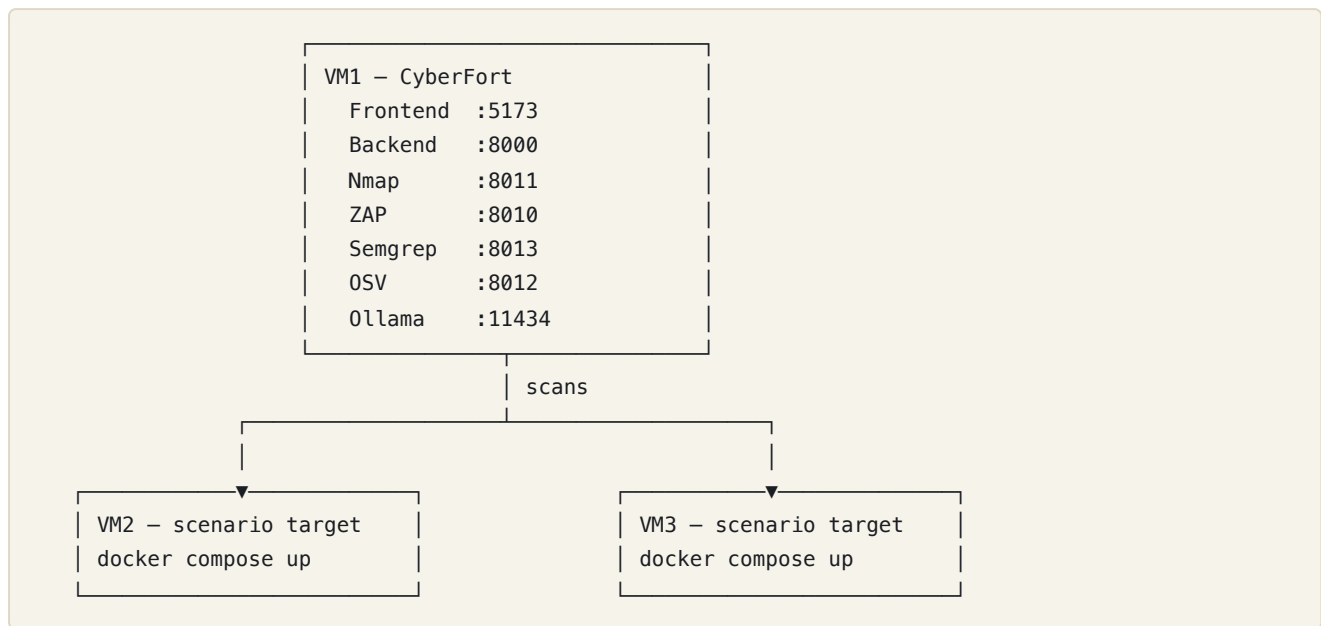


CyberFort cyber-range training scenarios

A set of three deliberately-vulnerable training exercises that showcase the **CyberFort** compliance platform on a three-VM cyber range, plus a **live pilot** run against a real commercial product. Each training scenario is from a **different sector** and is assessed against a **different compliance framework**, so they together demonstrate the full breadth of the platform; scenario 04 then proves it on a product that is actually going to market.

#	Folder	Sector	Compliance framework	Target stack	Primary CyberFort capabilities
1	01-portpilot-maritime/	Maritime	CRA (Cyber Resilience Act, Annex I + Vuln. Handling)	Flask + PostgreSQL	Nmap + ZAP + Semgrep + OSV + CRA conformity assessment + AI assistant
2	02-smartgrid-meter-energy/	Energy	NIS2 (Article 21(2) measures)	Flask admin + Mosquitto MQTT + Modbus TCP	Nmap (OT ports) + ZAP + Semgrep + OSV + NIS2 assessment
3	03-netlink-isp-digital-infra/	Digital infrastructure	ISO 27001:2022 (Annex A gap analysis)	Node.js Express + PostgreSQL	Nmap + ZAP + Semgrep (Node.js) + OSV (npm) + ISO 27001 assessment + JWT-forgery chain
4	04-clonesystems-seuxdr-siem/	ICT security vendor · pilot	CRA (full manufacturer conformity route, Annex III Class I)	Go + React SIEM/XDR with local-LLM remediation	SBOM Generator + OSV + Semgrep + VEX + CRA Technical File + conformity assessment + gap analysis + EU DoC readiness + CE marking checklist

Range topology



All three VMs sit on the same isolated training subnet. CyberFort reaches each scenario by IP. Scenarios run on either VM2 or VM3 depending on the day's session plan; only one scenario should be active per target VM at a time because their port mappings overlap (Postgres on 5432 in S1+S3).

What the scenarios show together

- **Four sectors** — maritime port operator, energy distribution operator, ISP, and an ICT security vendor. These mirror the SME profiles the CYBERFORT proposal explicitly targets.
- **Three frameworks** — CRA (product-side regulatory compliance), NIS2 (operator-side essential-entity obligations), ISO 27001 (general ISMS gap analysis). Each one is seeded in CyberFort with question texts the trainee actually answers.
- **Four technology stacks** — Python + Flask, Python + OT protocols (Modbus + MQTT), Node.js + EJS, and Go + React. Different scanner footprints (Semgrep covers Python, Node and Go; OSV covers PyPI, npm and Go modules).
- **One workflow** — register product → run scanners → convert findings to risks → answer the assessment → attach evidence → export PDF readiness report.
- **Two altitudes.** Scenarios 01–03 are two-hour *training* exercises against seeded targets: the trainee learns the workflow, and the findings are known in advance. Scenario 04 is a six-month *pilot* against a real product with unknown findings. It exercises the CRA-specific modules the training scenarios never touch — SBOM Generator, VEX statements, the six CRA Technical File pages, the EU Declaration of Conformity readiness score and the CE Marking Checklist — and then closes the loop: remediate the product, re-assess, and show the movement (7% → 70%). Run 01 first to learn the platform; run 04 to prove it.

How to use the scenarios

For each scenario the intended flow is:

1. Cyber range admin deploys the scenario stack on a target VM (`docker compose up -d --build` ; for scenario 04, `range/deploy.sh`).
2. Trainee logs into CyberFort on VM1.
3. Trainee follows `docs/TRAINEE_HANDBOOK.md` for that scenario — register the target as a product, run the relevant scanners, fill in the assessment, export the PDF readiness report.
4. Instructor reviews the PDF and the risk-register entries against `docs/INSTRUCTOR_GUIDE.md` .

Each scenario is sized for **two hours** of trainee time at intermediate level.

Repo layout

```
cyber-range-scenarios/
├─ README.md                ← this index
├─ scripts/
│   └─ md_to_pdf.py         ← markdown → PDF helper used to render the docs
├─ 01-portpilot-maritime/   ← Maritime / CRA / Flask
├─ 02-smartgrid-meter-energy/ ← Energy / NIS2 / Flask + MQTT + Modbus
├─ 03-netlink-isp-digital-infra/ ← Digital infra / ISO 27001 / Node.js
└─ 04-clonesystems-seuxdr-siem/ ← ICT vendor / CRA pilot / Go + React (real product)
```

Scenarios 01–03 follow the same structure: `README.md` , `docker-compose.yml` , application source, `docs/` (handbook + instructor guide + control mapping). Scenario 04 carries no application source — its target is a shipping commercial product in its own repository — so instead of a `docker-compose.yml` it ships a `range/` deployment kit that fetches the product at a pinned reference and adapts it for a range VM, in either a baseline or a remediated mode. Its `docs/` set is `PILOT_RUNBOOK.md` , `CRA_CONFORMITY_MAPPING.md` , `ASSESSOR_GUIDE.md` , `REMIEDIATION_BACKLOG.md` and `PILOT_TIMELINE.md` , plus the pilot presentation under `presentation/` . PDFs of all the markdown docs ship next to their `.md` counterparts and are regenerated via `python3 scripts/md_to_pdf.py <file>.md` .

Safety

Scenarios 01–03 are intentionally vulnerable. Scenario 04's target is a real product whose remediation actions delete files, kill processes and rewrite host firewall rules as root. **Run all four only on an isolated cyber-range network** — never on the public internet or on a corporate LAN.

Status

- ✓ Scenario 01 (PortPilot Maritime, CRA): complete, smoke-tested, PDFs rendered.
- ✓ Scenario 02 (SmartGrid Meter Energy, NIS2): complete, smoke-tested, PDFs rendered.
- ✓ Scenario 03 (NetLink ISP Digital Infrastructure, ISO 27001): complete, smoke-tested, PDFs rendered.
- ✓ Scenario 04 (Clone Systems SEUXDR, CRA pilot): complete, and closed the loop over a six-month engagement (Feb–Jul 2026). Baseline 2 of 30 CRA objectives compliant (2026-07-29, `t4.4_siem_ai_remediation @ 00a95ad`); remediation sprint 1 committed 2026-07-30 (`c4ba16b` , `cbd092b` , 60 files, 8,261 lines); re-assessed at 21 of 30 — **7% → 70%**. Screenshots captured from the live `CRA Extended` tenant; 27-slide pilot deck rendered.